

Section : Biology

Q.1 Given below are some stages of human evolution. Arrange them in correct sequence (Past to Recent)

- (A) *Homo habilis*
- (B) *Homo sapiens*
- (C) *Homo neanderthalensis*
- (D) *Homo erectus*

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (A), (B), (C), (D)
3. (D), (A), (B), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Match List-I with List-II

List-I

List-II

Terms

Features

- | | |
|----------------|--|
| (A) Allele | (I) The position on the chromosome where a gene occurs |
| (B) Locus | (II) Testing a suspected heterozygote by crossing it with a known homozygous recessive |
| (C) Gene | (III) One of two or more alternative nucleotide sequences at a single gene locus |
| (D) Test Cross | (IV) A length of DNA that codes for a particular polypeptide or protein |

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (D), (I), (B) - (II), (C) - (IV), D - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Match List-I with List-II

List-I

List-II

Abbreviation used in reproductive health

Features

- | | |
|----------|---|
| (A) IVF | (I) Sperm is directly injected into the ovum |
| (B) ZIFT | (II) Transfer of ovum from a donor into the fallopian tube of recipient |
| (C) GIFT | (III) Embryos with more than 8 blastomeres is transferred into the uterus |
| (D) ICSI | (IV) Fertilisation outside the body |

Choose the correct answer from the options given below:

1. (A)- (IV), (B)- (III), (C)- (II), (D)- (I)
2. (A)- (I), (B)- (II), (C)- (III), (D)- (IV)
3. (A)- (II), (B)- (I), (C)- (IV), (D)- (III)
4. (A)- (III), (B)- (IV), (C)- (I), (D)- (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Which one of the following statements is incorrect about ecological pyramids?

1. Pyramids of energy is always upright.
2. Saprophytes are given top place in ecological pyramids.
3. Pyramids of number can be inverted in the terrestrial ecosystem.
4. It doesn't take into account the same species belonging to two or more trophic levels.

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Which one of the following hormones is produced only during pregnancy?

1. hCG
2. Thyroxine
3. Cortisol
4. Estrogens

Options 1. 1

2. 2

3. 3

4. 4

Q.6 The condition where one species is harmed whereas the other is unaffected is called.

1. Amensalism
2. Parasitism
3. Predation
4. Competition

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Which of the following argument supports the conservation of biodiversity from the below statement:

"Waking up to a Bulbul's song in the morning"

1. Narrowly Utilitarian Argument
2. Broadly Utilitarian Argument
3. Ethical Argument
4. Philosophical Argument

Options 1. 1

2. 2

3. 3

4. 4

Q.8 Arrange the steps involved in creating pest-resistant transgenic plants using RNA interference (RNAi).

(A) Binding of complementary RNAs to form double-stranded RNA (dsRNA).

(B) Formation of sense and antisense RNA in the host cells.

(C) Silencing of specific mRNA of the nematode.

(D) Introduction of nematode-specific genes into the host plant using *Agrobacterium* vectors.

Choose the correct answer from the options given below:

1. (D), (B), (C), (A)
2. (D), (C), (B), (A)
3. (B), (A), (D), (C)
4. (D), (B), (A), (C)

Options 1. 1

2. 2

3. 3

4. 4

- Q.9** AIDS is a sexually transmitted disease (STD). Find out the correct statement with reference to it
- (A) It is caused by HIV belonging to the retrovirus group.
 - (B) It can pass through the placenta from an infected mother to her child.
 - (C) It can be diagnosed using ELISA.
 - (D) HIV enters into the B-lymphocytes and reproduces to produce progeny of virus.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.10** The term "biodiversity" was popularised by:

- 1. Paul Ehrlich
- 2. Gause
- 3. Edward Wilson
- 4. Ramdeo Mishra

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.11** Match List-I with List-II

List-I (Disorders)

List-II (Features)

- (A) Colour blindness (I) 45 with 'XO'
- (B) Haemophilia (II) Trisomy of 21st chromosome
- (C) Down's Syndrome (III) Mutation of certain genes on 'X' chromosome
- (D) Turner's Syndrome (IV) Protein involved in clotting of blood

Choose the correct answer from the options given below:

- 1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- 2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- 4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.12** Which one of the following diseases is non-infectious?

- 1. Cholera
- 2. Pneumonia
- 3. Tuberculosis
- 4. Elephantiasis

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.13** Arrange the following events of the menstrual cycle in the correct sequence:

- (A) The rising level of FSH stimulates the growth of the ovarian follicle and secretion of estrogen.
- (B) The corpus luteum secretes progesterone to maintain the endometrium.
- (C) The rapid secretion of LH leads to the rupturing of the Graafian follicle.
- (D) The release of unfertilized secondary oocyte leads to the menstrual cycle.

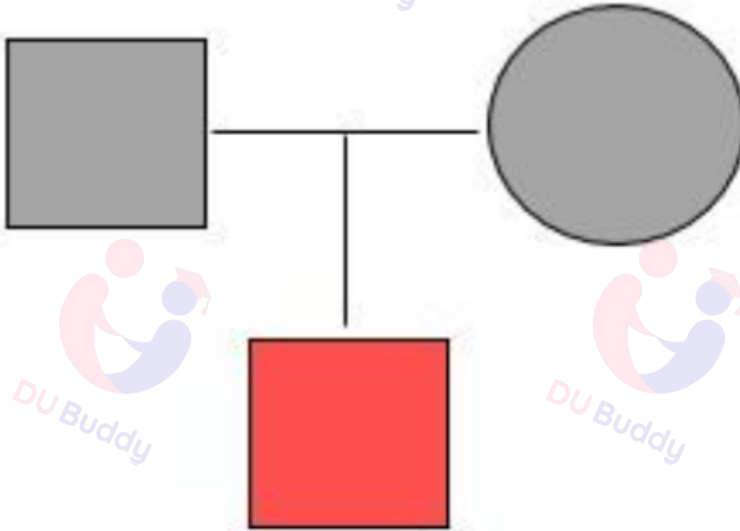
Choose the correct answer from the options given below:

1. (D), (A), (C), (B)
2. (D), (C), (B), (A)
3. (D), (A), (B), (C)
4. (D), (B), (C), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Study the human pedigree chart given below and choose the correct conclusion.



1. Parents with male child affected with disease
2. Parents with female child affected with disease
3. Normal male child with disease affected parents
4. Parents with normal male child

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Identify the incorrect statement about the human genome -

1. Repeated sequences make up very small portion of the human genome.
2. The human genome contains 3164.7 million bp.
3. Chromosome 1 has most genes (2968), and the Y has the fewest (231).
4. Less than 2 per cent of the genome codes for proteins.

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Which one of the following sex determination systems is found in honey bee?

1. XY type
2. Z and W type
3. Haplodiploid type
4. XO type sex

Options 1. 1

2. 2
3. 3
4. 4

Q.17 During allergic reactions, which of the followings are secreted?

- (A) Histamines
- (B) Interferons

(C) Serotonin

(D) Pyrogens

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (C) only
3. (A), (B), (C) and (D)
4. (A) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Which one of the following is an incorrect statement about endosperm production during double fertilization in plants?

1. Endosperm is produced when one of the two male gametes fuses with the nucleus of the female egg.
2. Endosperm is produced when one of the two male gametes fuses with the two polar nuclei located in the central cell to form a triploidy nucleus.
3. Endosperm tissues are filled with reserve food material and used for nutrition.
4. Endosperm of groundnut is completely consumed before seed maturation.

Options 1. 1

2. 2

3. 3

4. 4

Q.19 We should not allow mosquitoes to breed in our surroundings because they

1. Multiply very fast and cause pollution
2. Are vectors for many diseases
3. Bite and cause skin diseases
4. Are not important insects

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Match List-I with List-II

List-I (Terms)

(A) Divergent evolution

(B) Convergent evolution

(C) Evolution by anthropogenic action

(D) Adaptive radiation

List-II (Examples)

(I) Eye of the octopus and mammals

(II) Forelimbs in vertebrate animals

(III) Australian marsupials

(IV) Industrial melanization in moth

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.21 The pollination which involves the transfer of pollen grain from the anther to the stigma of another flower of the same plant is known as-

1. Geitonogamy
2. Xenogamy
3. Cleistogamy
4. Dichogamy

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.22 Transfer of genetic message from sequences of nitrogenous bases to sequences of amino acids is called -

- 1. Translation
- 2. Replication
- 3. Transcription
- 4. Transformation

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.23 Increase in population is due to:

- 1. Emigration and Mortality
- 2. Natality and Immigration
- 3. Immigration and Emigration
- 4. Mortality and Natality

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.24 Despite occupying about 70% of the surface area of earth, the productivity of the oceans is low because-

- 1. Nutrient availability is low at the bottom of the ocean as less sunlight penetrates into it.
- 2. Marine diversity is highly concentrated around coastal areas of the ocean.
- 3. High salinity inhibits plant growth in the ocean.
- 4. Low temperature at the bottom prevents plant and animal growth.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.25 Which one of the followings is major concern related to biopiracy?

- 1. Rich biodiversity and traditional knowledge are being exploited without compensation
- 2. Government providing compensatory payments for bio-resources
- 3. Ethical use of bio-resources for traditional medicine
- 4. Developed nations are developing bio-resources legally

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.26 Which of the following species of animals are now extinct:

- (A) Dodo from Mauritius
- (B) Thylacine from Australia
- (C) Quagga from Africa
- (D) Steller's sea cow from Russia

Choose the correct answer from the options given below:

- 1. (A), (B), (C) and (D)
- 2. (A), (B) and (D) only
- 3. (A), (B), and (D) only
- 4. (A), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.27 Identify the correct statements for pollen grains

- (A) Pollen grain represents male gametophyte.
- (B) Pollen grain is a haploid unicellular body.
- (C) The hard exine of the pollen is made up of sporopollenin.
- (D) The intine of the pollen is composed of oxidative polymers of carotenoids or carotenoid esters.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.28 HIV attacks which one of the following cells?

- 1. T-cytotoxic cells
- 2. T-helper cells
- 3. B-cells
- 4. T-memory cells

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.29 What is the role of the C peptide in insulin?

- 1. It helps insulin to bind receptors.
- 2. It is removed during maturation and helps in the formation of functional insulin.
- 3. It enhances the activity of mature insulin.
- 4. It prevents the synthesis of insulin in the pancreas.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.30 Cuckoo layings it's egg in the nest of crow for incubation is an example of-

- 1. Ideal Parasitism
- 2. Brood Parasitism
- 3. Fake Parasitism
- 4. Gaussian Parasitism

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.31 Copper T is used as an Intra Uterine Device (IUD) for contraception. Identify the correct statement for its mode of action

- (A) It is made up of ionized copper.
- (B) It releases Cu ions that suppress sperm motility.
- (C) It makes the uterus unsuitable for implantation.
- (D) It decreases sperm phagocytosis within the uterus.

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.32 Which of the following is not used to draw the ecological pyramid?

- 1. Fresh weight

2. Dry weight
3. No. of individuals
4. Rate of transfer of energy

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which one of the following biological products was produced by the transgenic cow named Rosie?

1. Human insulin
2. Human alpha-lactalbumin in milk
3. Human growth hormone
4. Human antibodies

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Genetically modified organisms have been useful in many ways. Which one of the following statements is not correct in the case of GMO plants?

1. GMO plants are more tolerant to abiotic stresses.
2. GMO plants rely on chemical pesticides.
3. GMO plants reduce post harvest losses.
4. GMO plants enhance nutritional value of food.

Options 1. 1

2. 2
3. 3
4. 4

Q.35 DNA fingerprinting techniques have the following steps. Arrange these steps in the correct order.

- (A) Digestion by restriction enzyme
- (B) Transferring separated DNA fragments
- (C) DNA-DNA hybridization
- (D) Gel Electrophoresis

Choose the correct answer from the options given below

1. (A), (D), (B), (C)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Identify the correct statement for Zygote Intra-Fallopian Transfer (ZIFT).

1. The *in-vitro* technique involving the transfer of an early embryo into the fallopian tube of the same female from whom the egg was taken.
2. The *in-vitro* technique involving the transfer of an ovum collected from a donor into the fallopian tube of another female.
3. The *in-vivo* technique involving the transfer of an early embryo into the fallopian tube of the same female from whom the egg was taken.
4. The *in-vivo* technique involving the transfer of an ovum collected from a donor into the fallopian tube of another female.

Options 1. 1

2. 2
3. 3
4. 4

Q.37 During the process of spermatogenesis, the secondary spermatocytes undergo _____ division to produce _____ spermatids.

1. Mitotic, haploid
2. Mitotic, diploid

- 3. Meiotic, diploid
- 4. Meiotic, haploid

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.38 The 'Roquefort cheese' is ripened by growing a specific _____ on them, which gives them a particular flavour.

- 1. Fungi
- 2. Yeast
- 3. Bacteria
- 4. Algae

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.39 Match List-I with List-II

List-I (Terms)

(A) IgE

(B) IgA

(C) Hepatitis B vaccine

(D) Polymorpho- nuclear leucocytes

List-II (Features)

(I) Formed by yeast

(II) Cellular barrier

(III) Antibodies due to allergens

(IV) Present in colostrum

Choose the correct answer from the options given below:

- 1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- 2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 The Convention on Biological Diversity held in Rio de Janeiro in 1992 is popularly known as

- 1. The Earth Summit
- 2. Stockholm Conference
- 3. Rio-20
- 4. The World Summit on Sustainable Development

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 Read the passage carefully and answer the given questions.

Recombinant DNA Technology

Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridization procedures used in plants and animals breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing genes into the target organism.

In Recombinant DNA Technology, what is the role of the "selectable marker gene"?

- 1. To provide energy to the host cell
- 2. To cut the foreign gene from the donor organism
- 3. To help in identification and selection of the host cells that contain the recombinant DNA
- 4. To promote the expression of the foreign gene

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the given questions.

Recombinant DNA Technology

Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridization procedures used in plants and animals breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing genes into the target organism.

What is the purpose of using a gene promoter in recombinant DNA technology?

1. To cut DNA at specific sequences
2. To insert the gene into the host organism's genome
3. To ensure the foreign gene expression in the host organism
4. To replicate the gene in a bacterial cell

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the given questions.

Recombinant DNA Technology

Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridization procedures used in plants and animals breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing genes into the target organism.

Which one of the following techniques is used to amplify a specific DNA sequence?

1. Polymerase Chain Reaction (PCR)
2. Gel Electrophoresis
3. Northern Blotting
4. DNA Sequencing

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the passage carefully and answer the given questions.

Recombinant DNA Technology

Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridization procedures used in plants and animals breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing genes into the target organism.

Which one of the following is not used as a vector in recombinant DNA technology?

1. Plasmid
2. Viruses
3. Bacterial chromosomes
4. Cosmids

Options 1. 1

2. 2

3. 3

Q.45 Read the passage carefully and answer the given questions.

Recombinant DNA Technology

Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridization procedures used in plants and animals breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing genes into the target organism.

Which one of the following options is correct for working of restriction enzymes?

1. A tool used to insert DNA into the cells
2. Enzyme used to join DNA fragments
3. Used as vector
4. Cuts DNA at specific sequence

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the given questions.

DNA Fingerprinting

Polymorphism in DNA sequence is the basis of genetic mapping of the human genome as well as of DNA fingerprinting. DNA fingerprinting involves identification of differences in some specific regions in the DNA sequence called repetitive DNA. It works on the principle of polymorphisms in DNA sequences. The repetitive DNA are separated from bulk genomic DNA. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA. These repetitive/satellite DNA sequences show a high degree of polymorphisms and form the basis of DNA fingerprinting.

DNA fingerprinting is meant for:

1. Slow way to compare DNA sequences
2. Quick way to compare DNA sequences
3. Identifying similarities in some specific regions in DNA
4. Identify mutations

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the given questions.

DNA Fingerprinting

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One of the most frequently used techniques in DNA fingerprinting is

1. VNTR
2. SCAR
3. SSCP
4. AFLP

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the given questions.

DNA Fingerprinting

Polymorphism in DNA sequence is the basis of genetic mapping of the human genome as well as of DNA fingerprinting. DNA fingerprinting involves identification of differences in some specific regions in the DNA sequence called repetitive DNA. It works on the principle of polymorphisms in DNA sequences. The repetitive DNA are separated from bulk genomic DNA. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA. These repetitive/satellite DNA sequences show a high degree of polymorphisms and form the basis of DNA fingerprinting.

Which one of the following techniques is involved in DNA fingerprinting?

1. Gel electrophoresis
2. Chromatography
3. Recombinant DNA Technology
4. ELISA

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer the given questions.

DNA Fingerprinting

Polymorphism in DNA sequence is the basis of genetic mapping of the human genome as well as of DNA fingerprinting. DNA fingerprinting involves identification of differences in some specific regions in the DNA sequence called repetitive DNA. It works on the principle of polymorphisms in DNA sequences. The repetitive DNA are separated from bulk genomic DNA. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA. These repetitive/satellite DNA sequences show a high degree of polymorphisms and form the basis of DNA fingerprinting.

DNA polymorphism forms the basis of

1. Transcription
2. DNA replication
3. Translation
4. Genetic mapping and DNA fingerprinting.

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the passage carefully and answer the given questions.

DNA Fingerprinting

Polymorphism in DNA sequence is the basis of genetic mapping of the human genome as well as of DNA fingerprinting. DNA fingerprinting involves identification of differences in some specific regions in the DNA sequence called repetitive DNA. It works on the principle of polymorphisms in DNA sequences. The repetitive DNA are separated from bulk genomic DNA. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA. These repetitive/satellite DNA sequences show a high degree of polymorphisms and form the basis of DNA fingerprinting.

Variable Number of Tandem Repeats are -

1. Satellite DNA
2. Silent DNA
3. SNPs
4. ESTs

Options 1. 1

2. 2

3. 3

4. 4